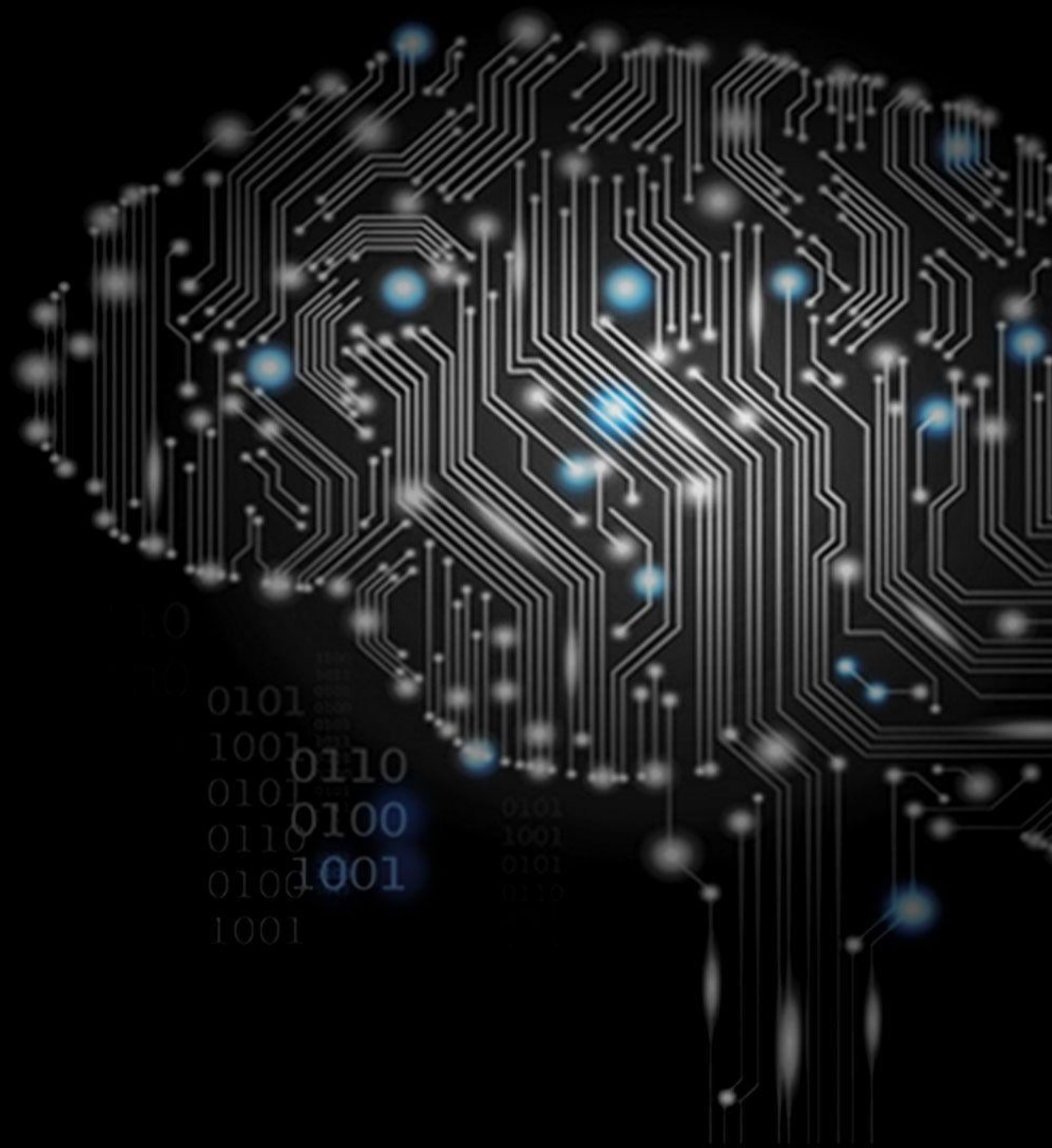


# Introduction to Generative AI

## 1. Foundations of Generative AI



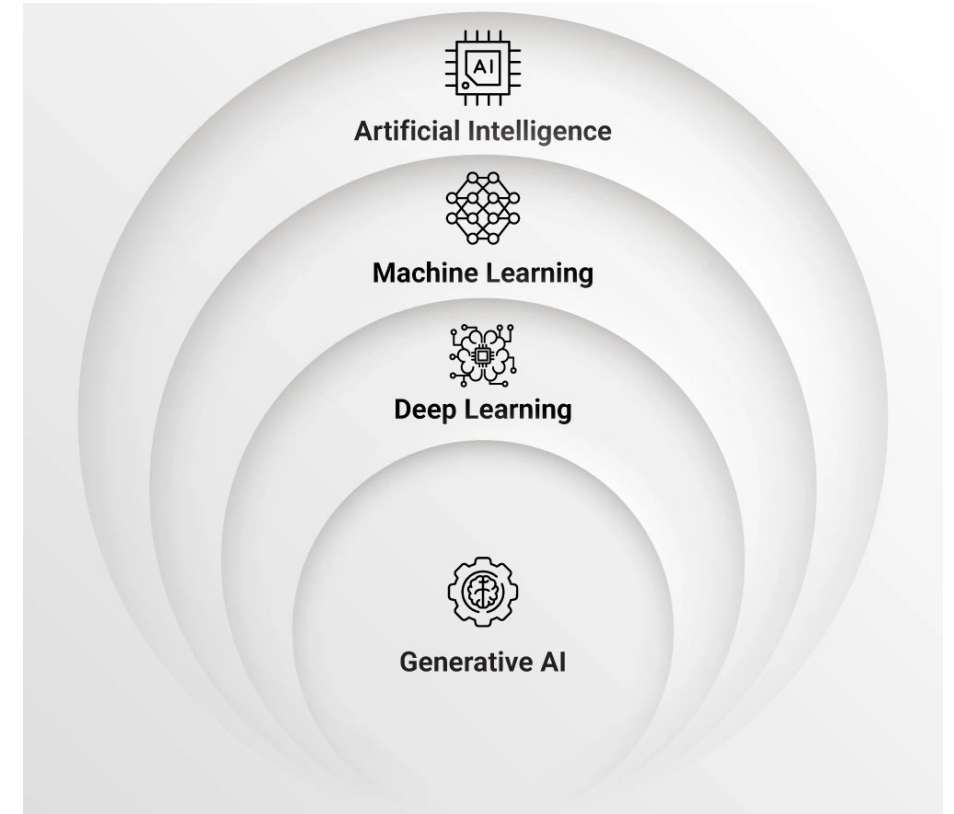
# Main Topics

- It is critical to separate out three related topics in order to understand the current state of generative AI and its potential organizational use
- AI technology
  - Generative AI and LLMs are just one small subset of the field of AI
  - AI technology is evolving and changing at a rapid pace
- AI business
  - Most of the current discussions around AI focus on the business of AI
    - *For example, the AI bubble that is of concern in the financial world*
    - *The impact of projects like data centres on the economy and environment*
    - *The strong push back by the public on what the LLM industry is doing*
- AI applications
  - How do AI applications integrate into an organization?
  - How have organizations failed at adopting AI and how have other succeeded?



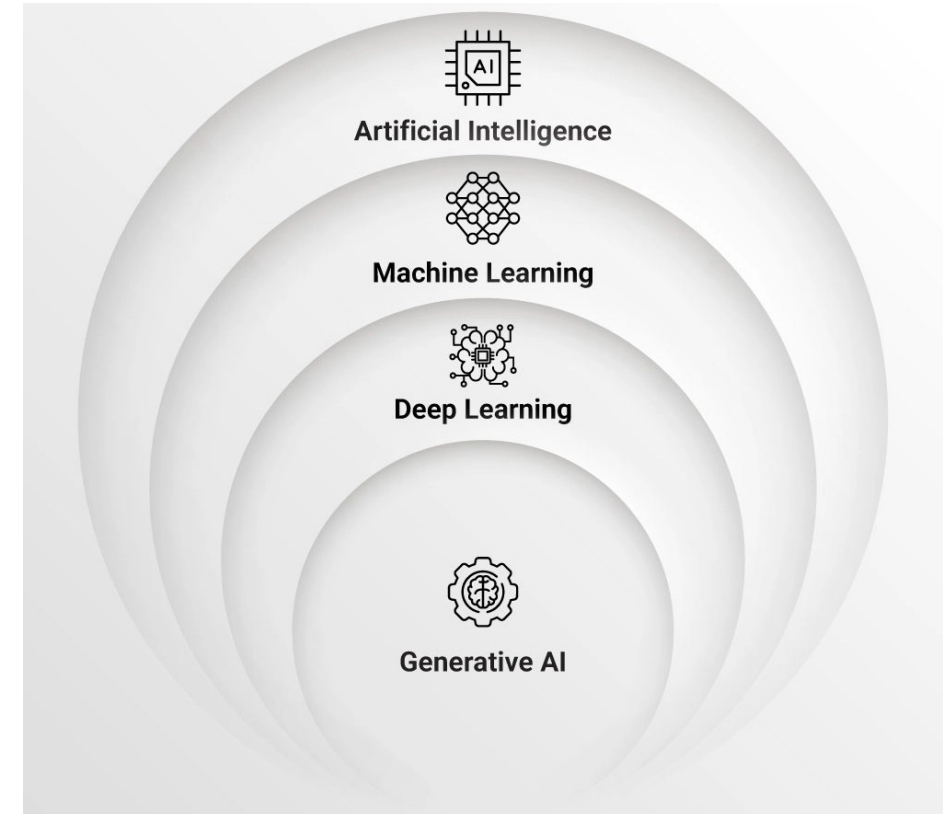
# Defining AI

- AI encompasses a number of different areas of study and types of applications
  - The current research and investment focus of AI is machine learning, which includes Large Language Models (LLMs) and Generative AI (GAI)
- There are other areas of AI that are not machine learning, for example
  - Cognitive computing
  - Artificial life and swarm intelligence
  - Evolutionary computing
  - Expert systems
  - Robotics and man/machine interfaces



# State of AI and ML

- In 2026, AI and ML is the hot economic topic
  - “The Algorithms” have become the modern bogey man
  - Or, alternatively, the technology that will produce a golden age
  - Most current development is in machine learning
- Implementation of ML has had mixed success
  - Often, when it works well, no one notices
  - Or when it works “good enough” like Siri or Google translate
  - People allow for some performance failures because expectations are low
  - However, when it fails, it can fail big and ugly
- Generative AI and LLMs are becoming disruptive
  - Deep fakes and compromising biometric authentication (eg. voice recognition)
  - Producing sophisticated and believable but incorrect results



# Success Story

## Deep Learning for Medical Diagnosis

Deep learning used to diagnose diabetic retinopathy

IBM

Medical Diagnosis

◀ PREV  NEXT ▶

Diabetic retinopathy is an eye disease that affects more than 126 million diabetics and accounts for more than 5% of blindness cases worldwide. Timely screening and diagnosis can help prevent vision loss for millions of diabetics worldwide, but many of them lack access to health care.

IBM and Julia Computing analyzed eye fundus images provided by Drishti Eye Hospitals, which provides eye diagnosis and care to thousands of rural Indians.

Drishti founder and CEO Kiran Anandampillai explains: “India is home to 62 million diabetics, many of whom live in rural areas with limited access to health facilities. Timely screening for changes in the retina can help get them to treatment and prevent vision loss. Julia Computing’s work using deep learning makes retinal screening an activity that can be performed by a trained technician using a low cost fundus camera.”





# Not so much



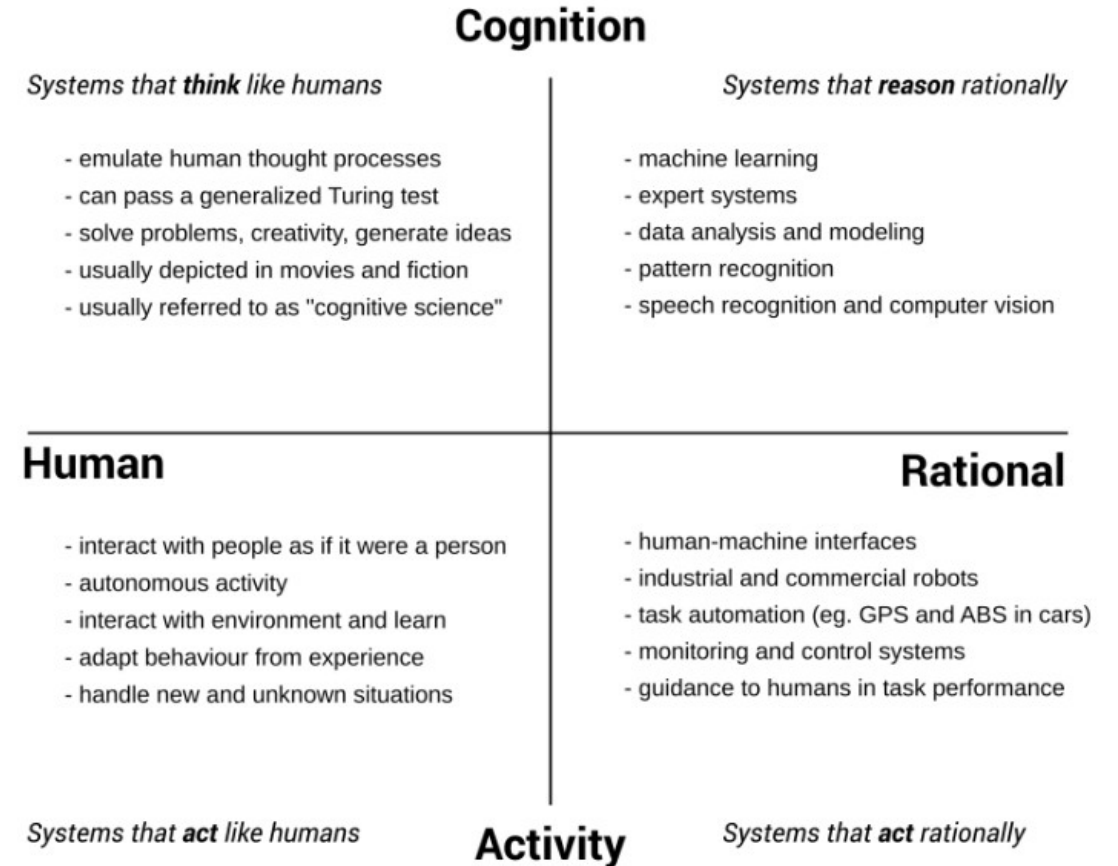
boonchai wedmakawand/Getty Images

- **A law firm was fined \$5,000 after one of its lawyers used ChatGPT to write a court brief.**
- **The document had included references to some cases and opinions that didn't exist.**
- **The lawyer said he had "no idea" ChatGPT could fabricate information.**



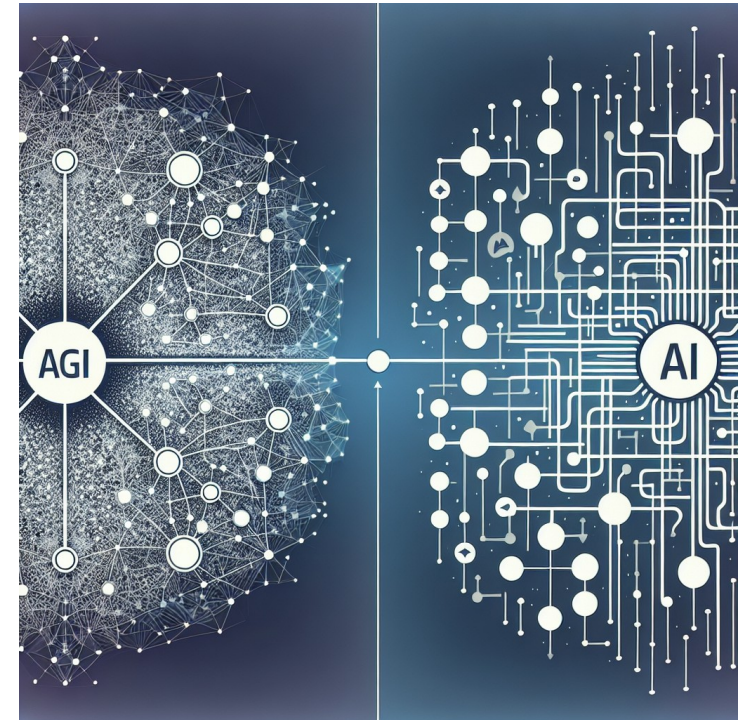
# The Norvig Matrix

- Provides a general classification of the different types of AI based on their goals
  - The “Human” column represents the goals of AGI, which are currently unattainable
    - *We do not understand human cognition and behaviour enough to simulate it*
    - *Posited that we have to “grow” AGI by raising it like a child*
  - The “Rational” column represents what we currently do in AI
    - *Systems that reason rationally produce results that are acceptable to humans, although the techniques used are not the same as people use*
    - *Systems that act rationally perform tasks in ways that are acceptable to humans but do not use the same processes as humans*



# AGI and ANI

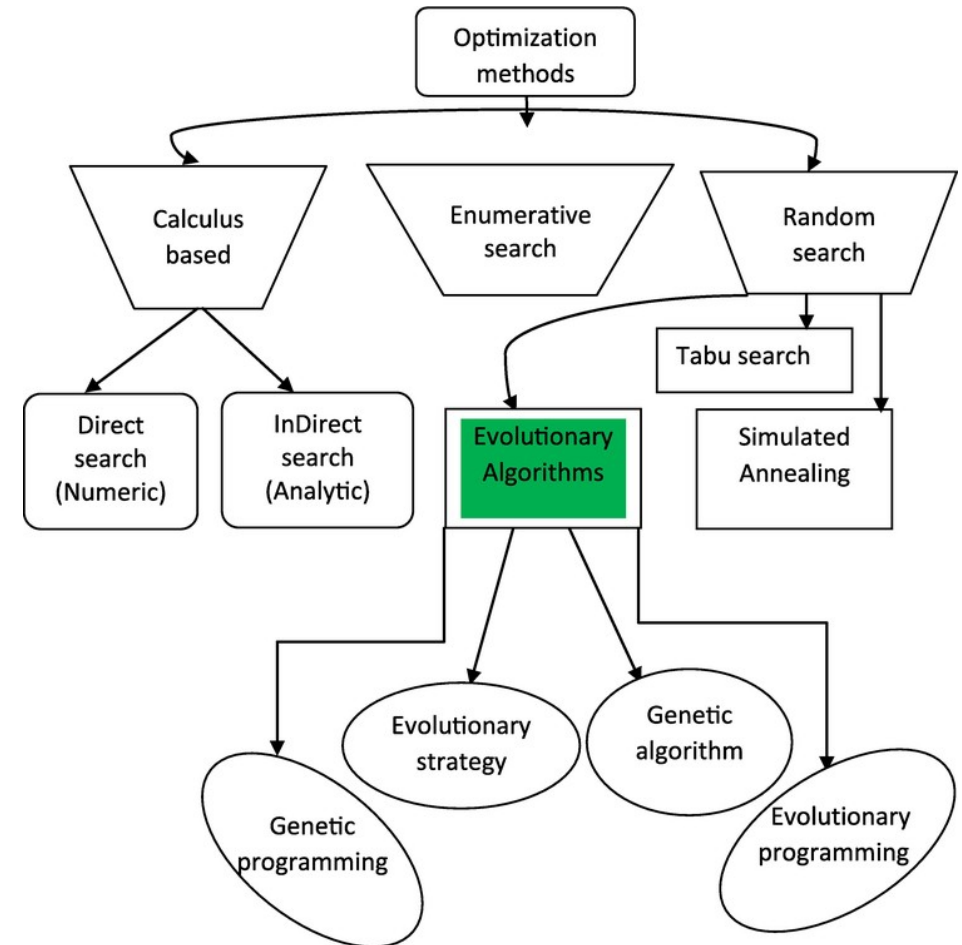
- Artificial general intelligence (AGI)
  - AI that possesses human-like intelligence and is capable of understanding and learning any intellectual task that a human being can
    - *Often referred to as “strong AI” or “full AI”*
    - *Able to perform a wide range of tasks and adapt to new situations without being explicitly programmed to do so*
    - *Beyond any current capability we have*
- Artificial narrow intelligence (ANI)
  - AI systems that are designed for specific tasks or functions.
    - *Specialized and limited in scope*
    - *Focusing on solving particular problems or performing specific functions within a narrow domain*





# Genetic and Evolutionary Computation

- Uses algorithms modelled after natural selection and evolution
  - Often used to solve problems that are too computationally expensive to solve otherwise
  - For example, PostgreSQL uses genetic computation to help optimize queries
- This area of AI focuses on developing optimal constraint based problems
  - For example, scheduling and resource allocation



# Evolutionary Example

- The image shows a lightweight, lattice-like structural component for an extra-planetary lander
  - The structure has an organic, webbed geometry that differs significantly from traditional rigid, boxy aerospace frames
  - This is characteristic of evolutionary design
  - It appears organic because evolutionary algorithms remove unnecessary material while reinforcing stress-bearing regions
  - Similar to how bones or natural forms evolve under environmental pressures



*NASA's Jet Propulsion Laboratory used generative design to develop a concept extraplanetary lander with lower mass and improved performance.*

# Swarm Intelligence Example

- Inspired by collective behavior in nature (ants, bees, birds, fish schools)
  - Intelligence emerges from many simple agents following local rules
  - Decentralized and self-organizing
  - Global behavior arises from local interactions
  - Highly adaptive, scalable, and robust while being tolerant to failure of individual agents
- Works well in complex, dynamic environments
  - Naturally suited for parallel computing
  - Scales efficiently with problem size
  - Useful when traditional optimization struggles
  - Aligns with decentralized AI and multi-agent systems

Ukraine has begun using AI-powered drone swarms in combat against Russian forces, marking a significant step in modern warfare technology. According to The Wall Street Journal [report](#), Ukrainian forces are leveraging software developed by local company Swarmer that allows groups of drones to coordinate, communicate, and adapt their actions independently of direct operator control.

## FROM EXPERIMENTAL TECH TO ROUTINE USE

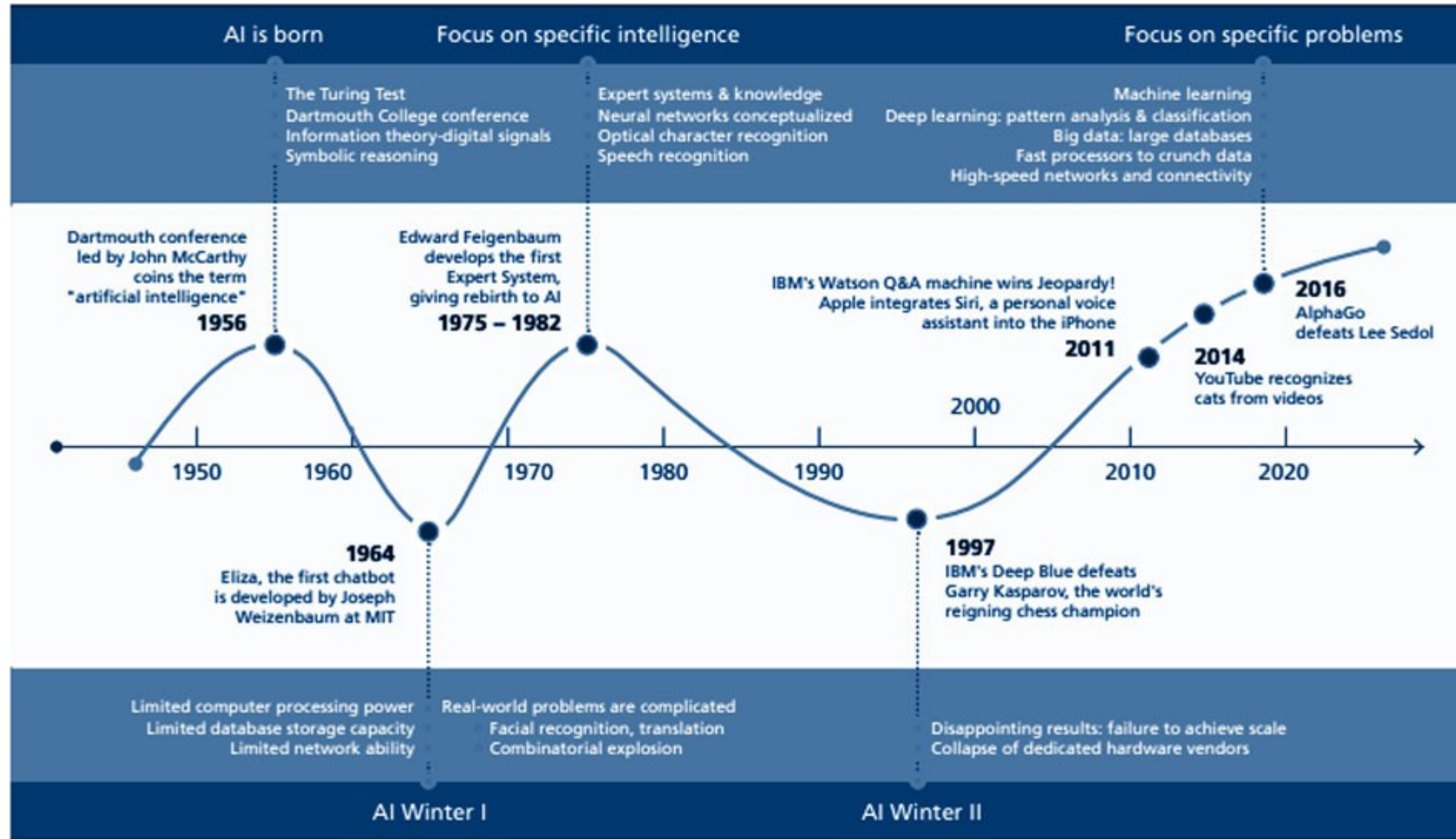
Military analysts say these are the first documented cases of swarm technology being used routinely in active combat. Ukrainian officials confirm the systems have been deployed for over a year, initially to lay mines and later to strike Russian troops, vehicles, and infrastructure.

**“You set the target and the drones do the rest. They work together, they adapt,” said Serhii Kuprienko, Swarmer’s chief executive.**

A typical operation involves one reconnaissance drone mapping routes for strike drones carrying small bombs. Once an enemy position is identified, the drones themselves decide which aircraft will release munitions and when. Ukrainian officers report the technology has been tested with as many as 25 drones, though Kuprienko said it is being scaled toward swarms of more than 100.



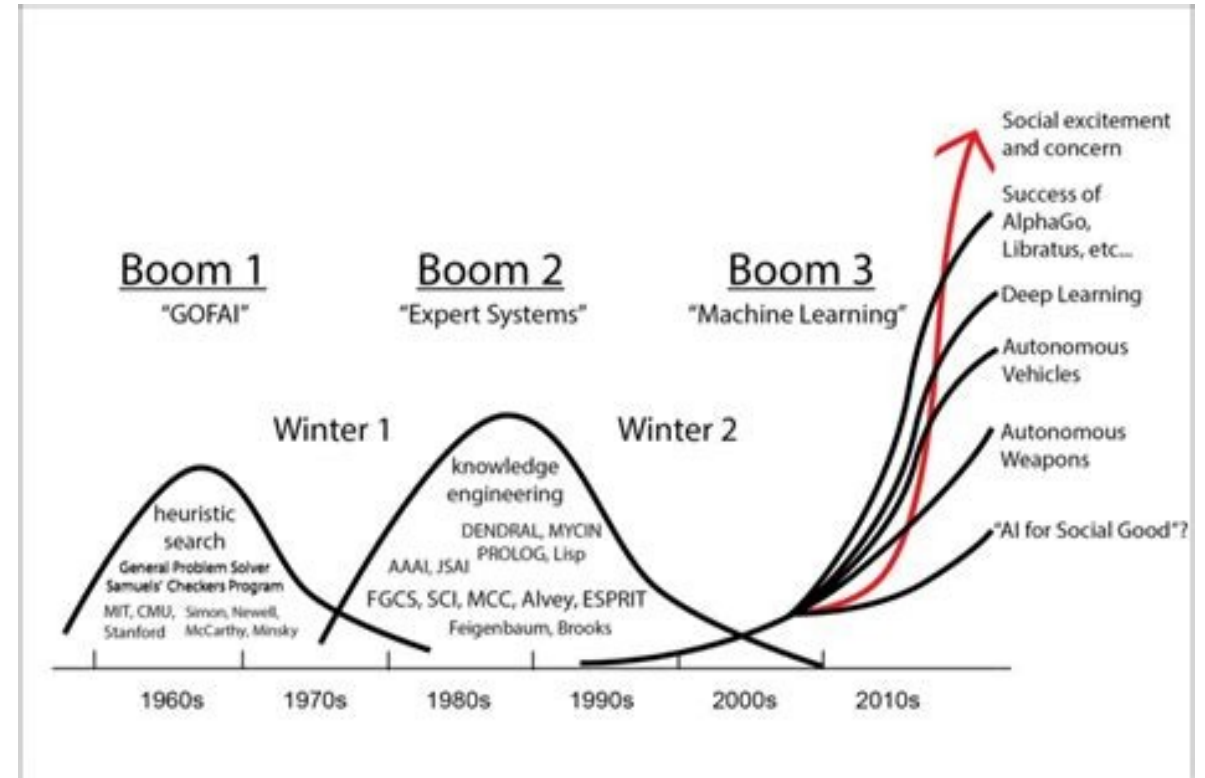
# AI Timeline





# AI Cycles

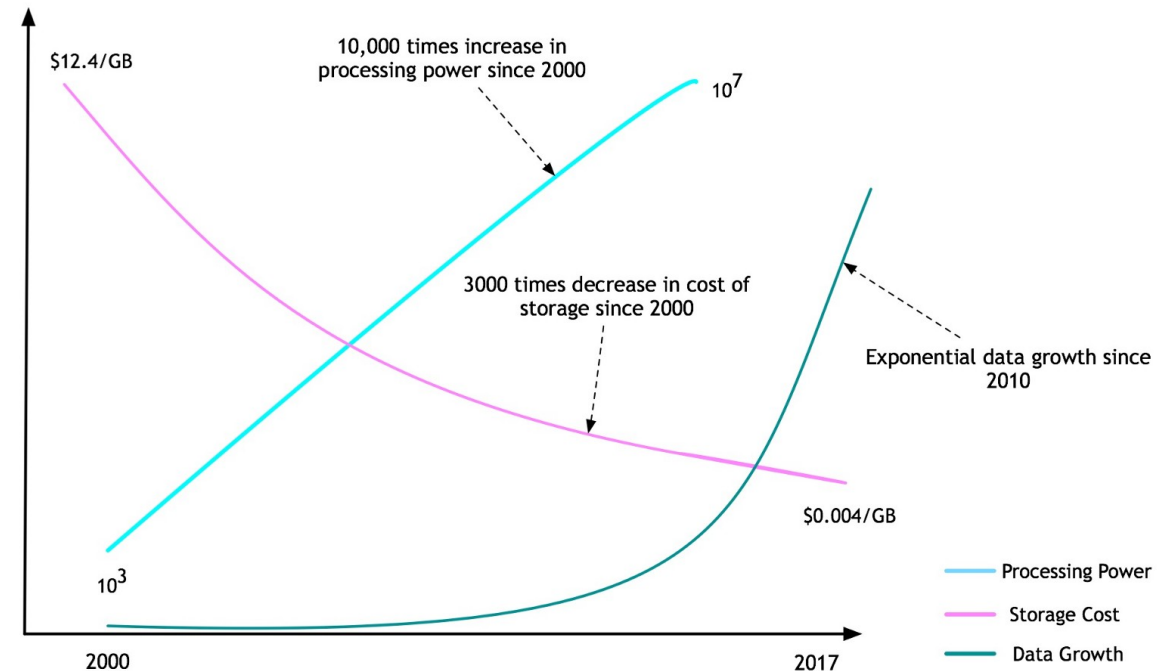
- AI has been around since the 1940s
- There have been three main AI booms
  - Enabled by advances in hardware and availability of compute resources
  - Supported by investment and funding
- Each was followed by an AI winter
  - AI companies were not able to deliver on their promises
  - Funding dried up
  - AI research continued during these winters but there was little commercial or industrial development





# Drivers of AI - Compute

- Many of the ideas of modern AI were developed as far back as 1947, neural networks for example
  - Implementing them was beyond the capability of computing capabilities
- The first requirement for AI to be successful is enough computing power to build the models and systems
  - This has been one of the main enablers of modern LLM and GAI development
- Current hot issue in LLMs is whether or not development can be sustained because of resource usage



# Drivers of AI - Theory

- The second driver is whether the theoretical applied mathematical tools are available
  - These are required to implement the ideas of AI into actual working models on a computing platform
- For example
  - Machine learning depends on the gradient descent algorithm
  - Training neural networks depends on the back propagation algorithm
  - LLMs could only be developed after Transformers were invented

Neural Networks Multi Layer Perceptrons

## Back Propagation Learning Algorithm

$$\frac{\partial E(\mathbf{w})}{\partial w_{ji}^{l-1}} = \sum_{i=1}^p \frac{\partial E(\mathbf{w})}{\partial y_j^{l-1}(i)} \frac{\partial y_j^{l-1}(i)}{\partial net_j^{l-1}(i)} \frac{\partial net_j^{l-1}(i)}{\partial w_{ji}^{l-1}}$$

$\frac{\partial E(\mathbf{w})}{\partial w_{ji}^{l-1}} = \sum_{i=1}^p \frac{\partial E(\mathbf{w})}{\partial y_k^l(i)} \frac{\partial y_k^l(i)}{\partial net_k^l(i)} \frac{\partial net_k^l(i)}{\partial y_j^{l-1}(i)} \frac{\partial y_j^{l-1}(i)}{\partial net_j^{l-1}(i)} \frac{\partial net_j^{l-1}(i)}{\partial w_{ji}^{l-1}}$

$-\delta_k^l(i)$

$f'(net_k^l(i))$   
 Depends on the activation function

$w_{kj}^l$

$f'(net_j^{l-1}(i))$   
 Depends on the activation function

$y_i^{l-2}(i)$

$$y_k^l(i) = f(net_k^l(i)) \quad y_j^{l-1}(i) = f(net_j^{l-1}(i))$$

$$net_k^l(i) = \sum_{j=1}^m w_{kj}^l y_j^{l-1}(i) \quad net_j^{l-1}(i) = \sum_{i=1}^n w_{ji}^{l-1} y_i^{l-2}(i)$$

President University Erwin Sitompul>NNFL 4/8

# Classical Machine Learning

- Classical ML refers to statistical models created using different algorithms
  - These are drawn from other fields like statistics and data analysis
  - ML introduced techniques to create these models using “training” algorithms that worked on large data sets
    - *Eg. The use of gradient descent to automatically converge to an answer*
    - *Eg. Measures to test how accurate and reliable the model was at making predictions*
- Generally applied to a single problem
  - Spam detection in email
  - Tumour detection in medicine

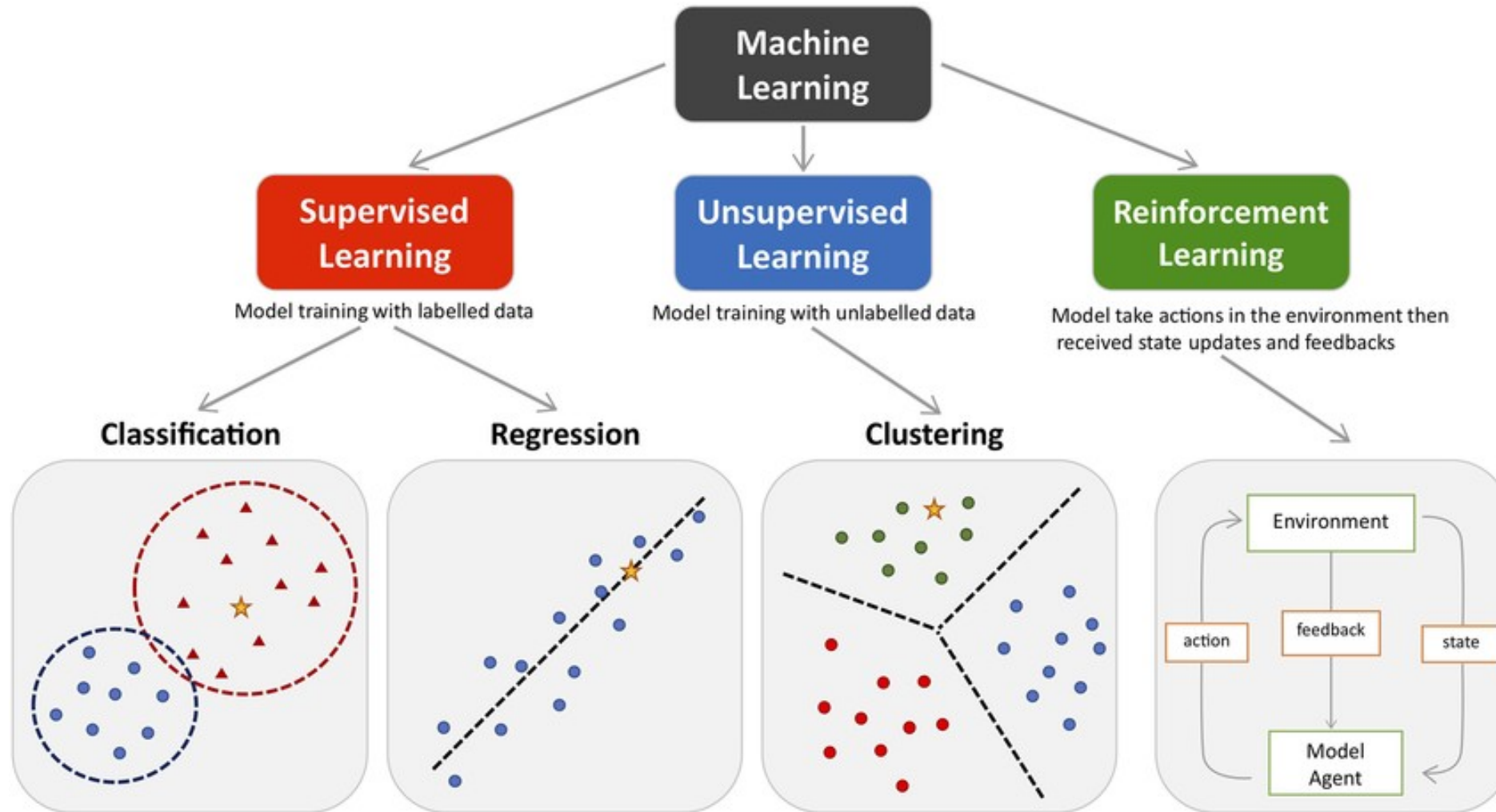


# Machine Learning Generations

- Generation One: Traditional ML
  - Standard supervised and unsupervised models
- Generation Two: Deep Learning
  - Use of neural nets for more complex problems
- Generation Three: LLMs
  - Introduction of Transformers
  - Large Language Models
- Generation Four: Generative AI
  - Uses LLMs to create new types of output
- Each generation extends the previous
  - Uses the tools and techniques of the previous generation
  - Stimulated by improvements in data, compute and algorithms



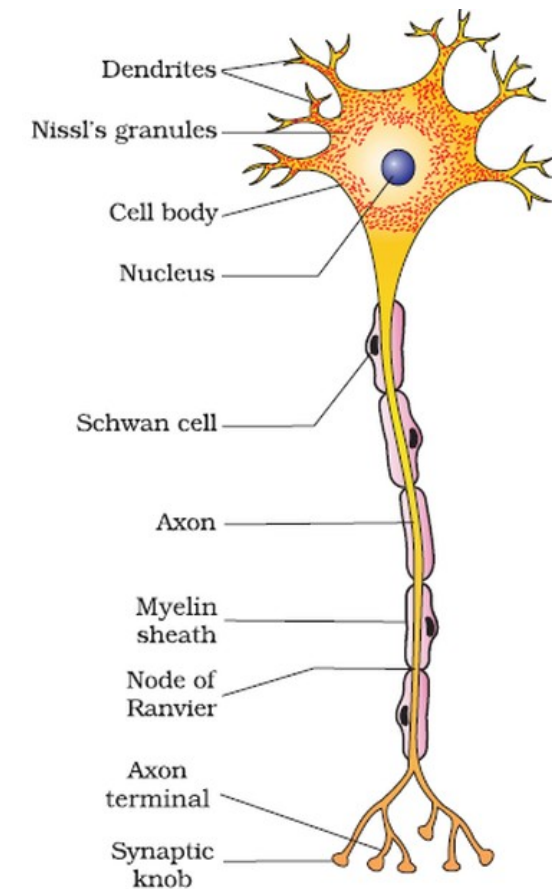
# Classical Machine Learning





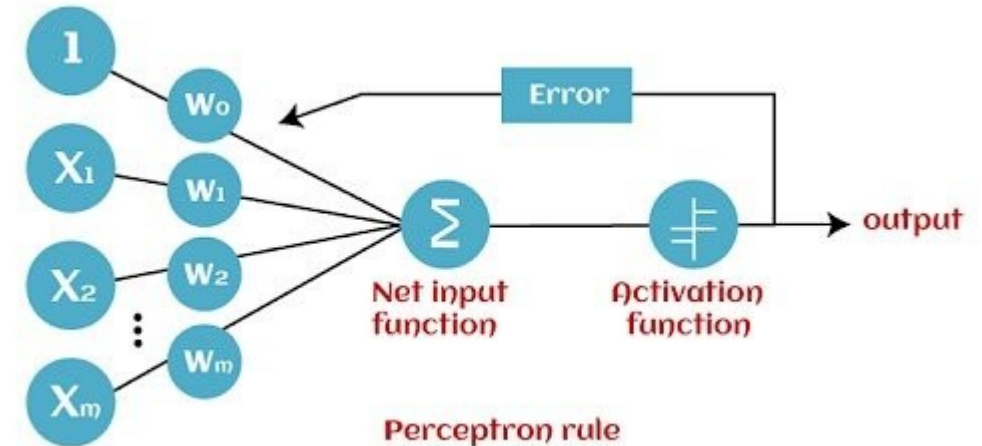
# Neural Networks

- Artificial neural nets (ANN) are at the core of Deep Learning
  - Powerful, scalable and can solve complex problems like classifying billions of images
  - ANNs inspired by neurons in human brain
  - Original theoretical formulation in the 1940s
- First working models in the 1960s
  - Very limited in power and could not simulate actual neurons
  - But early work provided an architectural model for deep learning
  - No longer intended to simulate human neurons



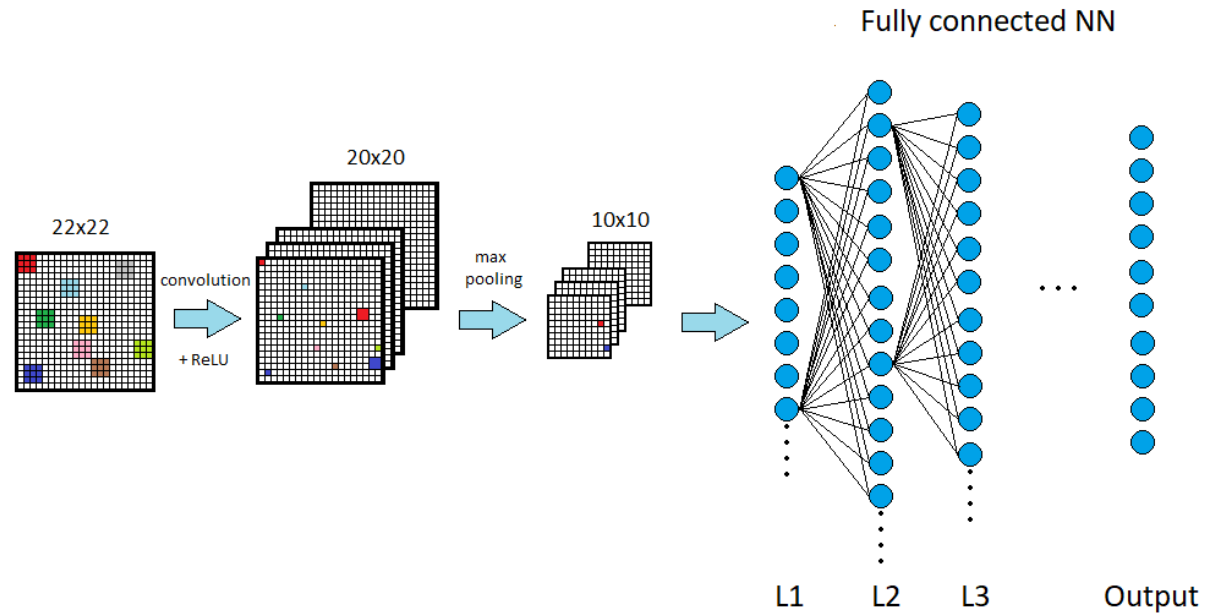
# Deep Learning

- Neural network
  - Refers to the organization and topology
  - Uses nodes and connections
  - Can be visualized as a graph
- Deep learning
  - Refers to the algorithms that execute on a neural network architecture
- Many deep learning applications use a combination of a specific NN architecture and a set of specific deep learning algorithms



# Convolutional NN

- CNNs were developed for image processing
- Imagine a small patch being slid across the input image
  - This sliding is called convolving
  - Similar to a flashlight moving from the top left end progressively scanning the entire image
  - This patch is called the filter/kernel and the area under the filter is the receptive field
  - The idea is to detect local features in a smaller section of the input space, section by section, to eventually cover the entire image
- These led to GAI image processing



# Problems with Text

- Text processing is more difficult than image processing
- To understand text, the system has to remember what it read previously
  - It has to remember the context of the discourse
- Needed for
  - Disambiguation of word meanings (was the screwdriver a drink or a tool?)
  - Anaphora resolution (who does “he” refer to?)
  - Basic understanding of what the text is about

Bob had always hated Dave. He always had.

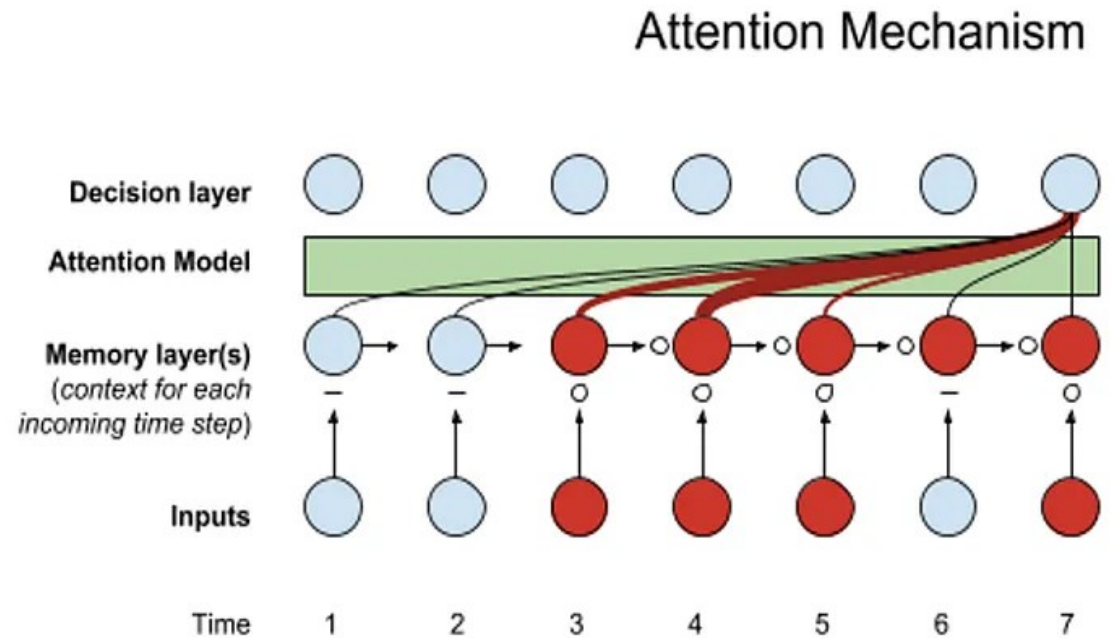
One day, he was drinking at his favourite bar when he walked in. Furious that he would show up at his favourite bar, he threw a screwdriver in his face.

Dave wasn't hurt so he declined to press charges.



# Attention

- When you read, not every word has the same importance
  - Focus is on certain key words that carry meaning
    - *“Although he was tired, John finished the report before the deadline.”*
    - *When trying to understand who finished the report, more attention is paid to “John” and “finished” and less to filler words like “although” or “the”*
  - Attention mechanisms help the model focus more on the important words when making a decision or prediction
    - *“Not all words matter equally, so pay more attention to these ones that do matter”*
  - It requires analyzing a sentence and its context at the same time looking for patterns of occurrence
  - And then developing relationships in the model to reflect the patterns in the text





# Transformers

- Transformers read an entire sentence, or paragraph, at once
  - They can figure out which words relate to each other, no matter where they appear
- “The bank approved the loan because the customer had excellent credit”
  - A transformer
    - Understands that “approved” is connected to “bank”
    - Connects “excellent credit” with the reason for approval
    - Doesn’t get confused if you rearrange the sentence slightly
    - Uses attention to weigh the importance of each word in relation to others

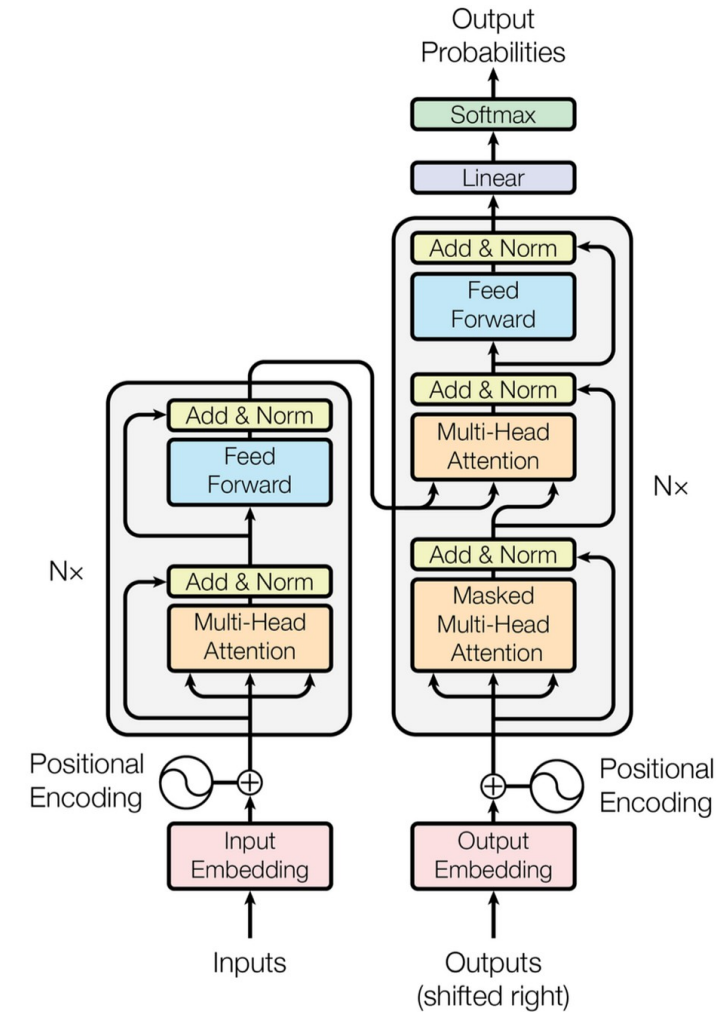
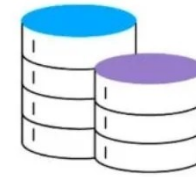


Figure 1: The Transformer - model architecture.

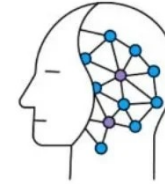
# Large Language Models

- Essentially a massive transformer based statistical model language
  - Trained on massive amounts of data
  - Has billions or trillions of dimensions
  - Takes massive compute power to train, usually using specialized hardware and parallel processing
- Transformers introduced
  - Self-attention
  - Parallel processing
  - Long-range dependency modeling
  - Efficient scaling

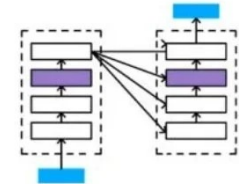
## Large Language Models (LLM)



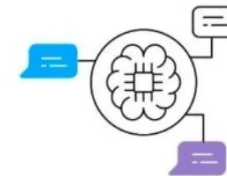
Massive Dataset



Deep Learning



Transformer Architecture



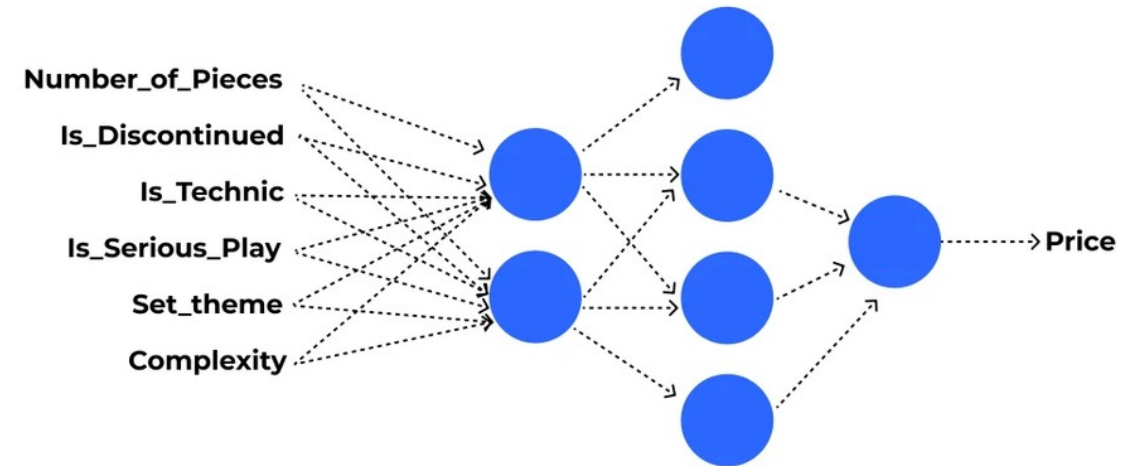
Self-supervised Learning



Fine-tuning

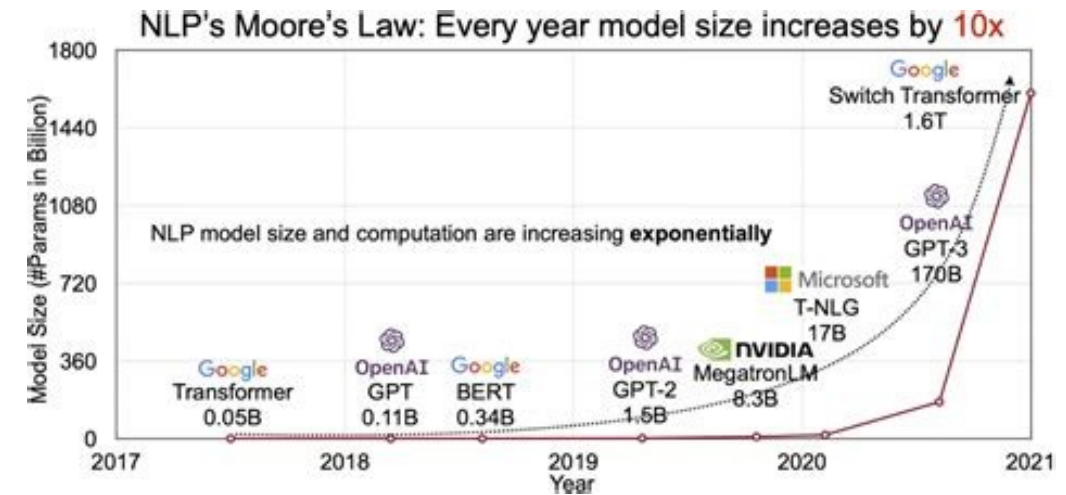
# Parameters

- A parameter or dimension
  - Internal variable of a model whose value is learned from training data, rather than set manually by the developer.
  - Define how the model transforms inputs into outputs
  - During training, an optimization algorithm adjusts the parameter so the model's predictions get closer to the true targets.
- Once training is complete
  - The parameters are fixed and represent what the model has "learned"
  - Saving a model essentially means saving its parameter values.
- The number of parameters is a rough measure of a model's capacity.



# LLM Size

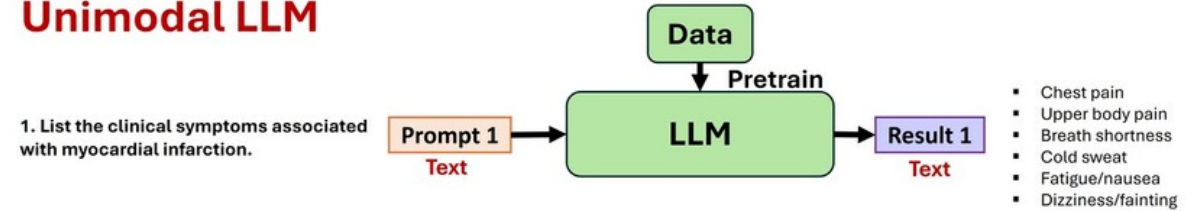
- A rough estimate of the size of an LLM is the number of parameters it trains on
  - Estimates for modern LLMs are in the order of trillions of parameters
  - The actual number is often kept secret by the model developers
- Diminishing returns
  - Larger models are not showing the expected improvements
  - Current interest is looking at other less monolithic architectures



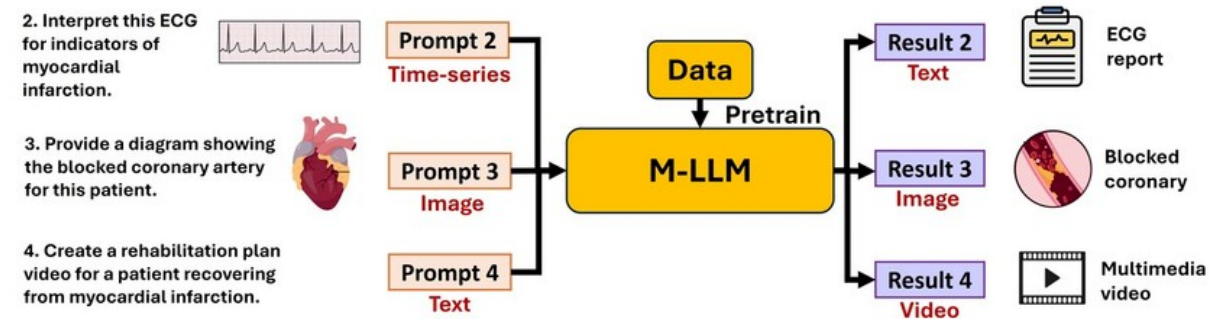
# Multi-modal Training

- Multi-modal means training a model on multiple types of data
  - Text, images, audio, or video
  - Enables processing cross different types of information
  - Use cases
    - *Image captioning*
    - *Visual question answering*
    - *Text-to-image generation*
    - *Document understanding (text + layout + images)*
    - *Voice assistants*
    - *Video understanding*
  - How It Works (Conceptually)
    - *Separate encoders for different modalities*
    - *Shared embedding space*
    - *Cross-modal attention mechanisms*
    - *Joint training objective*

## Unimodal LLM



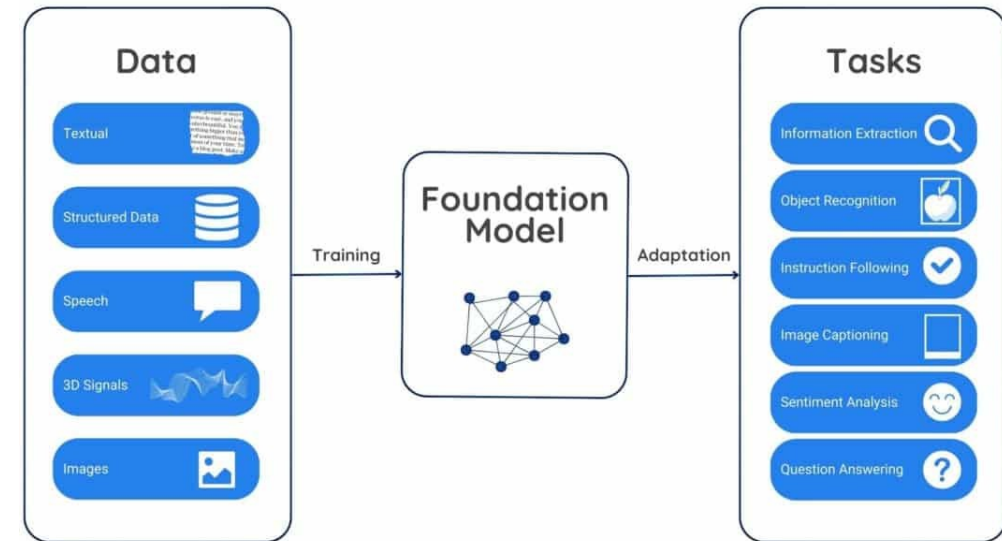
## Multimodal LLM





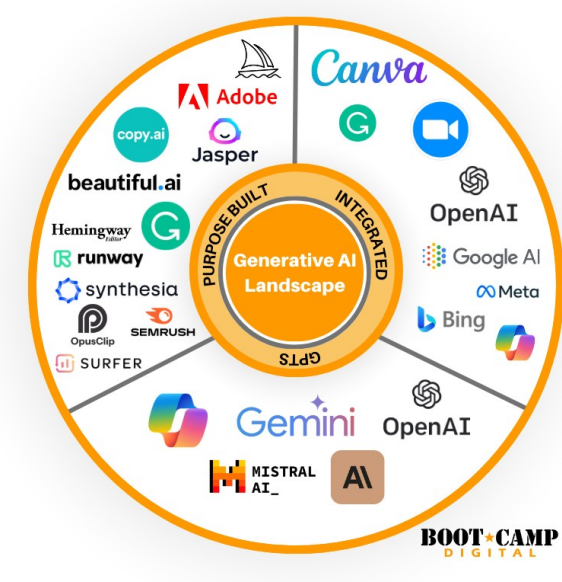
# Foundation Models

- A Foundation Model is a large-scale model trained on broad multi-modal data that can be adapted to many downstream tasks
- Basic characteristics
  - Trained on massive, diverse datasets
  - General-purpose rather than task-specific
  - Can be fine-tuned to adapt the model to different tasks or context
- Serves as a base (“foundation”) for many applications
  - The term LLM is often used when people are referring to a foundation model



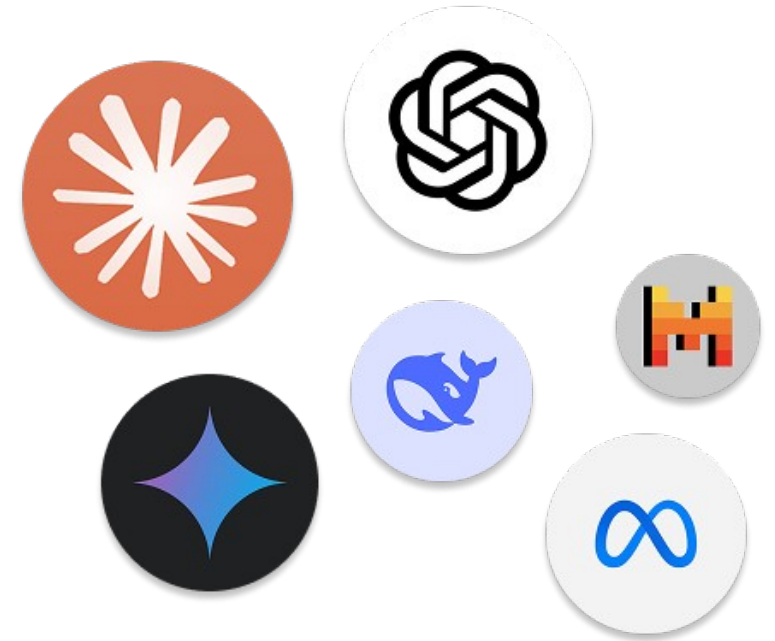
# The Generative AI Ecosystem

- Generative AI today is shaped by three major forces
  - Frontier model labs (training large foundation models)
  - Cloud and infrastructure providers (compute + distribution)
  - Open-source communities (democratization + rapid innovation)
- The ecosystem is highly interconnected:
  - Labs build models
  - Cloud providers distribute them
  - Open-source drives experimentation and adoption



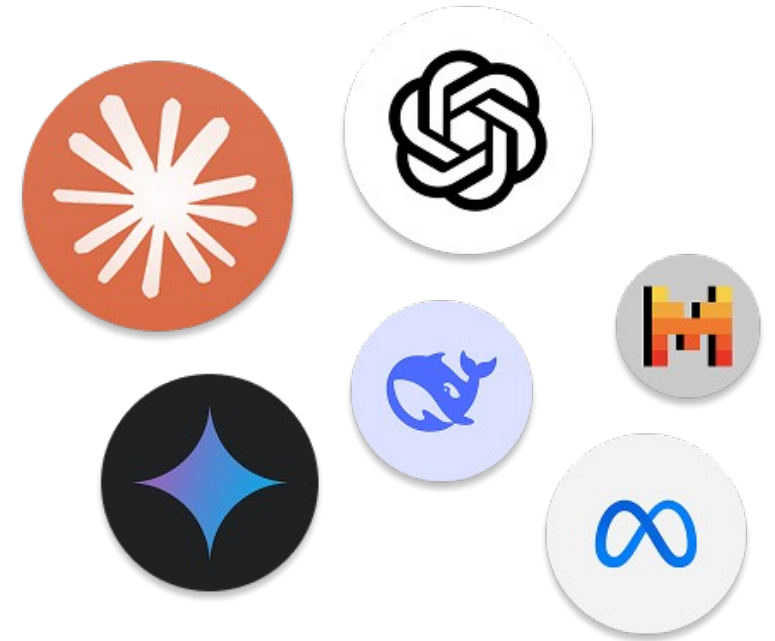
# The Generative AI Ecosystem

- Frontier model labs
  - Trains state-of-the-art foundation models
  - Operates at massive computational scale
  - Invests heavily in AI research and safety
  - Works with billions to trillions of parameters
  - Pushes new capabilities
  - Reasoning, multimodality, agents, etc
- Requires enormous capital and specialized infrastructure



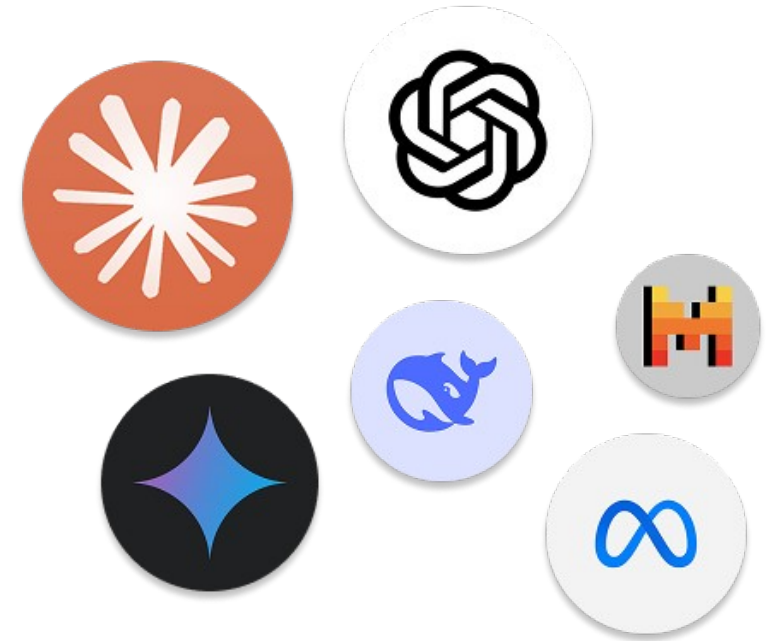
# The Generative AI Ecosystem

- Frontier labs operate where
  - Costs are extremely high
  - Technical uncertainty is high
  - Capabilities are advancing rapidly
- Examples
  - OpenAI
  - Anthropic
  - Google DeepMind
  - Meta (AI research division)
  - Mistral (emerging European lab)



# Why So Few Frontier Labs

- Training frontier models requires
  - Massive GPU clusters
  - Advanced distributed systems
  - Specialized ML researchers
  - Access to large-scale datasets
  - Billions of dollars in funding
- Only a small number of organizations can afford this.



# Commercial Model Providers

- OpenAI
  - Pioneer of GPT series (GPT-3, GPT-4, GPT-4o)
  - Popularized the LLM product model (ChatGPT)
- Strong focus on
  - Multimodal models
  - Tool usage
  - Agents
  - Deep partnership with Microsoft
- Business model: API + enterprise licensing
- Strategic role: Set early pace in scaling LLMs commercially.





# Commercial Model Providers

- Anthropic
  - Creator of Claude models
- Strong emphasis on
  - AI safety
  - Constitutional AI
  - Alignment research
- Competes directly with OpenAI in enterprise LLM space
  - Partnerships with Amazon (AWS Investment)
  - Strategic role: Safety-focused frontier model lab



# Commercial Model Providers

- Microsoft
  - Infrastructure provider (Azure AI)
  - Strategic investor in OpenAI
- Embeds LLMs into
  - Office
  - GitHub Copilot
  - Enterprise stack
  - Focus on AI as enterprise productivity layer
- Strategic role: Distribution + enterprise integration.



# Open-Source Ecosystem

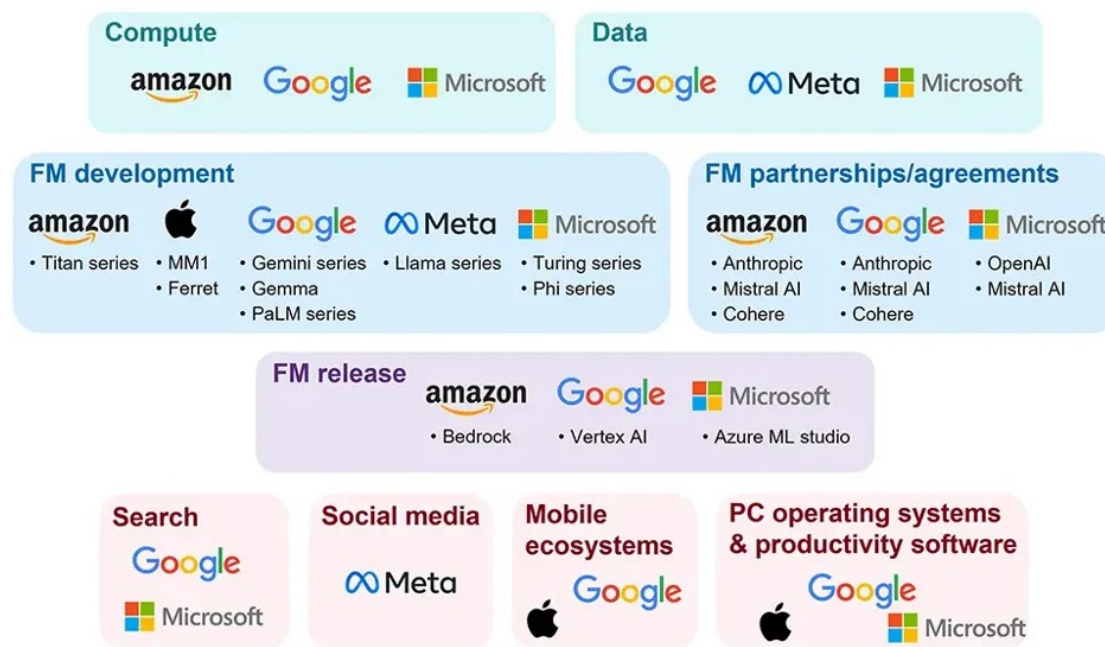
- Hugging Face
  - Central hub for open models
  - Model hosting + sharing platform
  - Transformers library
  - Datasets + evaluation tools
  - Community-driven experimentation
  - Strategic role: GitHub of AI models.els.



**Hugging Face**

# Industry Dominance

- The generative AI landscape is:
  - Oligopolistic at the frontier (few companies can train trillion-parameter models)
  - Highly competitive
  - Deeply dependent on cloud infrastructure
  - Increasingly shaped by safety, regulation, and geopolitics



# Discriminative AI

- Traditional ML and Deep Learning are discriminative
  - That means that they are trained on data to do one of two things
- Make decisions
  - Classify data into categories (or make predictions)
  - The model discriminates between possible choices
  - Often used to re-engineer a logical process that was once used but is no longer available
  - Or to emulate the result of a process that cannot be represented logically
    - *Eg. Replicate a banker's successful "gut feeling" for credit risk*
    - *Often experts make good decisions but they can't describe what they do so we emulate it with machine learning*
  - But we need to select the features of parameters to use to make the decision



# Discriminative AI

- Find relationships among data data points
  - Clustering data – find data that is near to each other using some measure of nearness
  - We have to determine which features will result in a useful clustering
  - The clustering is often a basis for making decisions

|                                                                                     |                                                                                     |                                                                                     |                                                                                     |                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|  |  |  |  |  |
| <b>Detached</b>                                                                     | <b>Disillusioned</b>                                                                | <b>Apathetic</b>                                                                    | <b>Passively Engaged</b>                                                            | <b>Actively Engaged</b>                                                             |
| <i>Politically uninvolved and unconcerned</i>                                       | <i>Politically sceptical and dissatisfied</i>                                       | <i>Uninterested in politics</i>                                                     | <i>Satisfied with the status quo</i>                                                | <i>Politically active and confident</i>                                             |
| ** Participate<br>✓ Vote<br>** Knowledge<br>** Discuss<br>□ Efficacy                | * Participate<br>✓ Vote<br>* Knowledge<br>□ Discuss<br>** Efficacy                  | * Participate<br>** Vote<br>* Knowledge<br>* Discuss<br>* Efficacy                  | ** Participate<br>✓ Vote<br>✓ Knowledge<br>✓ Discuss<br>✓✓ Efficacy                 | ✓✓ Participate<br>✓ Vote<br>✓✓ Knowledge<br>✓✓ Discuss<br>✓ Efficacy                |



# Generative AI

- Assuming we have a large enough training set and enough features
  - Each data point maps to a very large feature vector, maybe billions of features
  - Assume the data points have labels
  - Then we can generate a new data point from a set of features that are similar to other data points with that label
- Synthetic data
  - The use of GAI to create new data points that are similar to existing data points, x-rays of cancer tumors for example, is called synthetic data
  - Very useful for training models when not enough training data exists



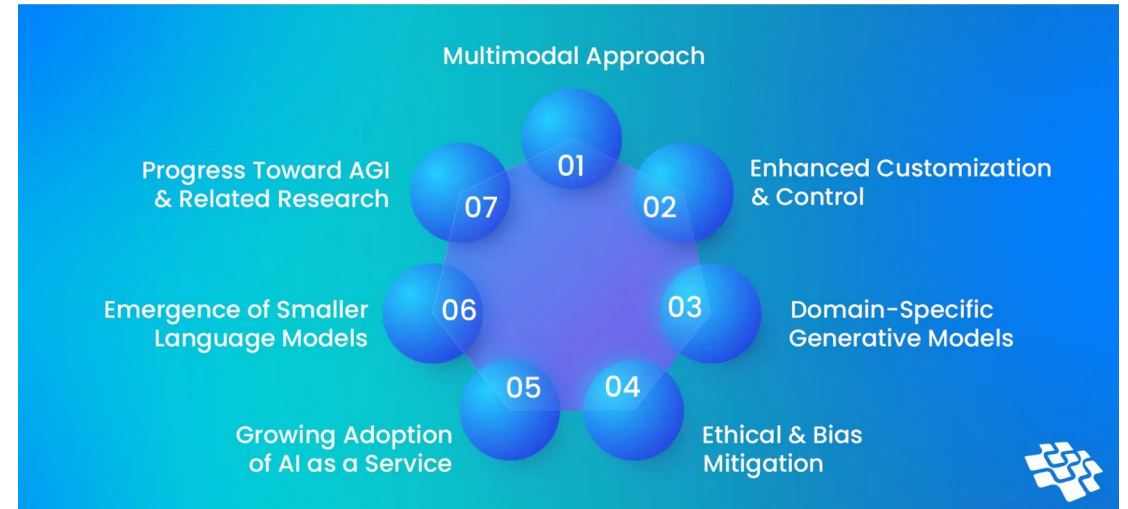
# Generative AI

- This allows for predictive capabilities
  - Given enough data in an input structure (often a sequence like a string of word) it allows for prediction of what comes next
  - This can also be used to generate text string or other sequences
- Transformational capabilities
  - Given two different types of data (images of faces and paintings by Picasso) it is possible to create a new data point with features common to both
  - “Your selfie as if it was painted by Picasso”
  - Create a photo of George Washington drinking wine



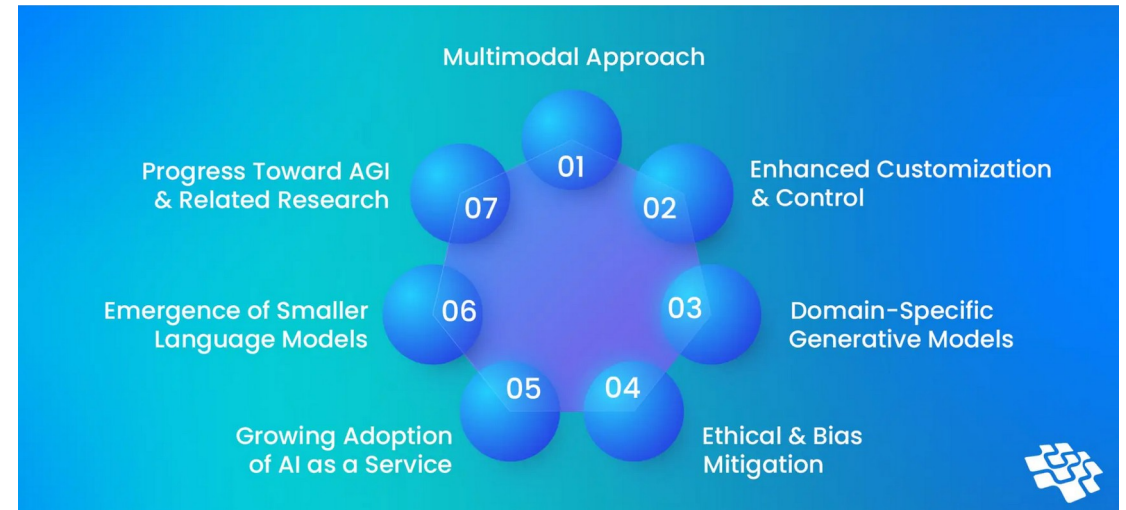
# Current Trends

- Smaller and more capable models
- Rapid improvement in
  - 3B–15B parameter models
  - Distilled frontier models
  - Quantized production-ready variants
  - Sparse / mixture-of-experts (MoE) inference optimization
- Performance gap between
  - 70B+ models
  - Well-tuned 7B–13B models
  - Is shrinking for many enterprise tasks



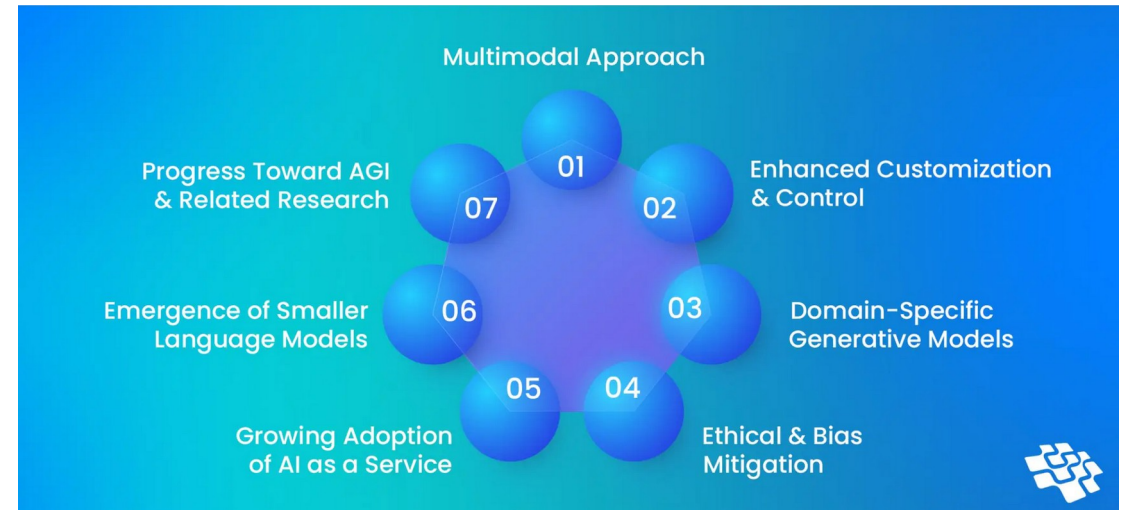
# Current Trends

- Native multimodal models
  - Multimodal models are becoming:
    - *Native (not stitched pipelines)*
    - *Better at cross-modal reasoning*
    - *More reliable in production*
  - Within 1–2 years
    - *Video understanding APIs will mature*
    - *Speech-to-reasoning-to-action pipelines will be standardized*
    - *Diagram-to-code workflows will improve significantly*



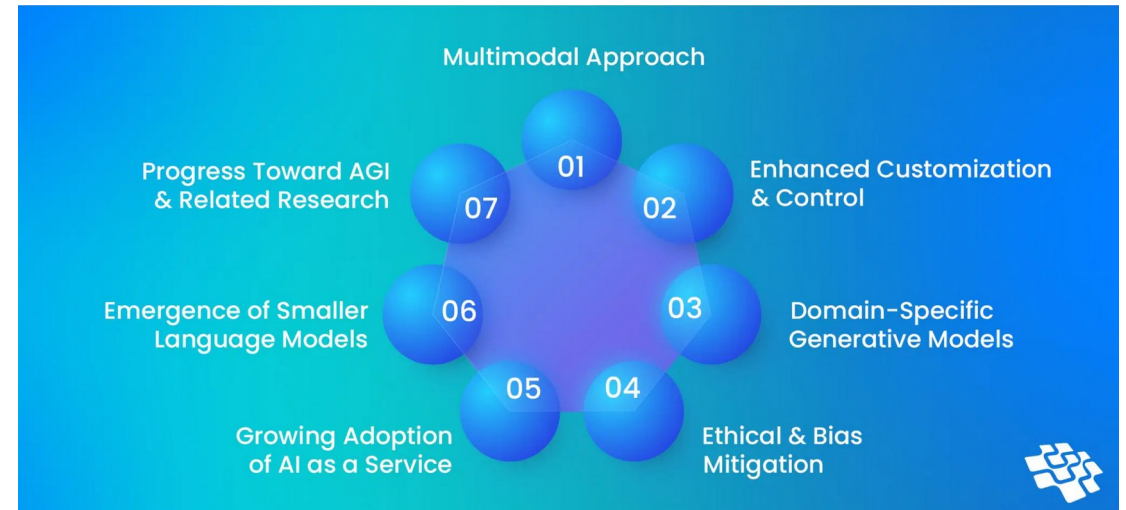
# Current Trends

- Structured output: tool first
  - LLMs are shifting from “Generate text” to “Generate structured actions”
  - Trends in LLMs
    - *Function calling as default*
    - *JSON schema enforcement*
    - *Tool routing layers*
    - *Deterministic execution wrappers*
  - Within 1–2 years
    - *Schema-constrained decoding to improve*
    - *Fewer malformed JSON outputs*
    - *Native validator-integrated decoding*
    - *LLMs become planners and not just executors*

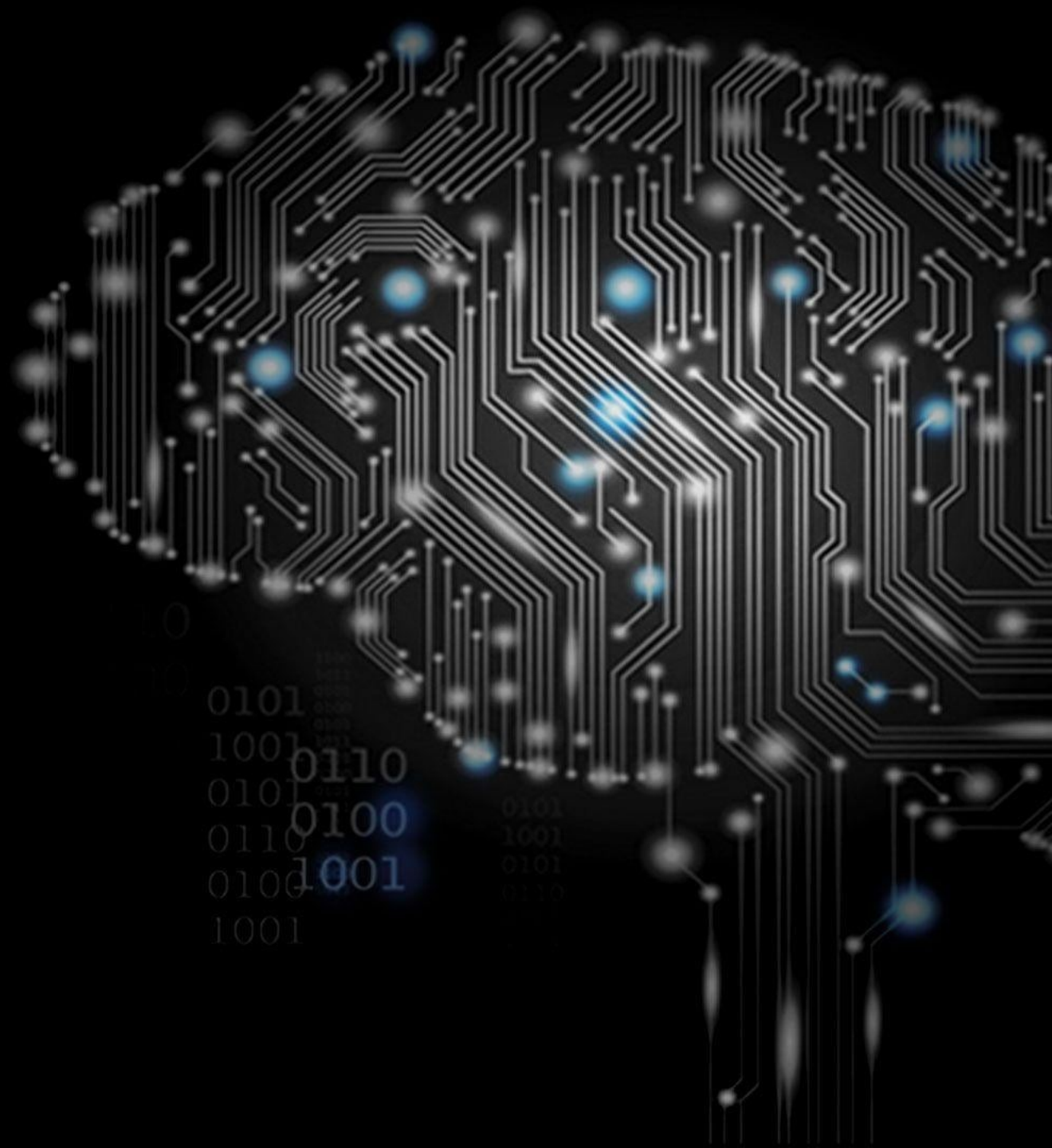


# Current Trends

- Agentic systems with guardrails
  - Agents are real but chaotic without control
  - Within 1–2 years
    - *Constrained agent frameworks*
    - *Hybrid reasoning + LLM systems*
    - *Not fully autonomous AGI-style agents*
    - *Controlled orchestration systems*
  - What will mature
    - *Multi-step reasoning pipelines*
    - *Long-horizon task execution with supervision*
    - *Retry and self-correction loops*
    - *Observability tooling for agents*







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